

COMPRESSED NN FOR DIABETES DETECTION ON FPGA WITH HLS₄ML

Vangos Theodoros 3765

Oikonomopoulos vasilis 3406

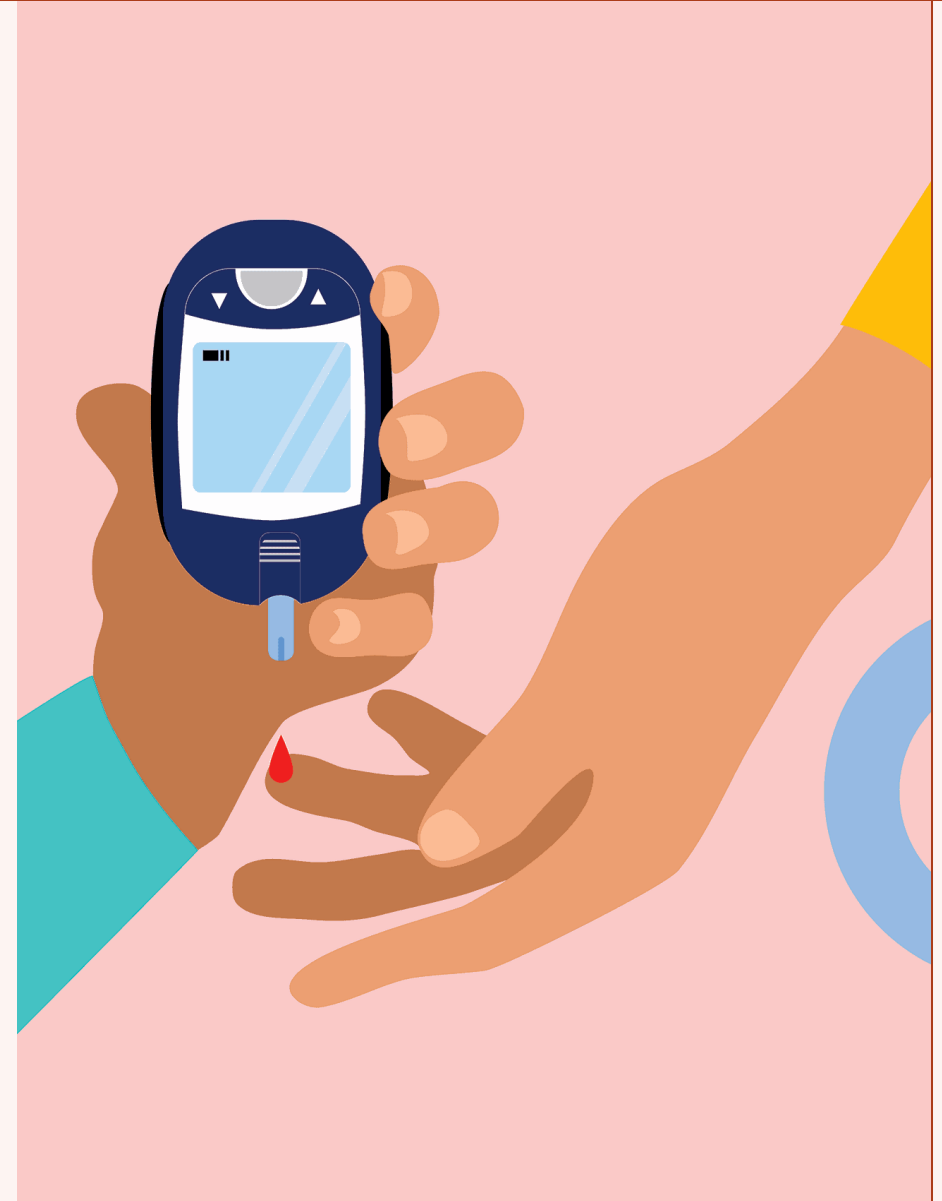


Diabetes detection

- Diabetes is a widespread chronic disease
- Machine Learning can identify relationships between clinical features
- Neural networks can perform binary classification from patient data

FPGA can offer:

- Low-latency execution of machine learning models
- Hardware acceleration
- Potential embedded AI applications



Our Goals:**1.Reduce:**

- Lantency
- LUTs
- FFs
- DSP usage
- Memory footprint

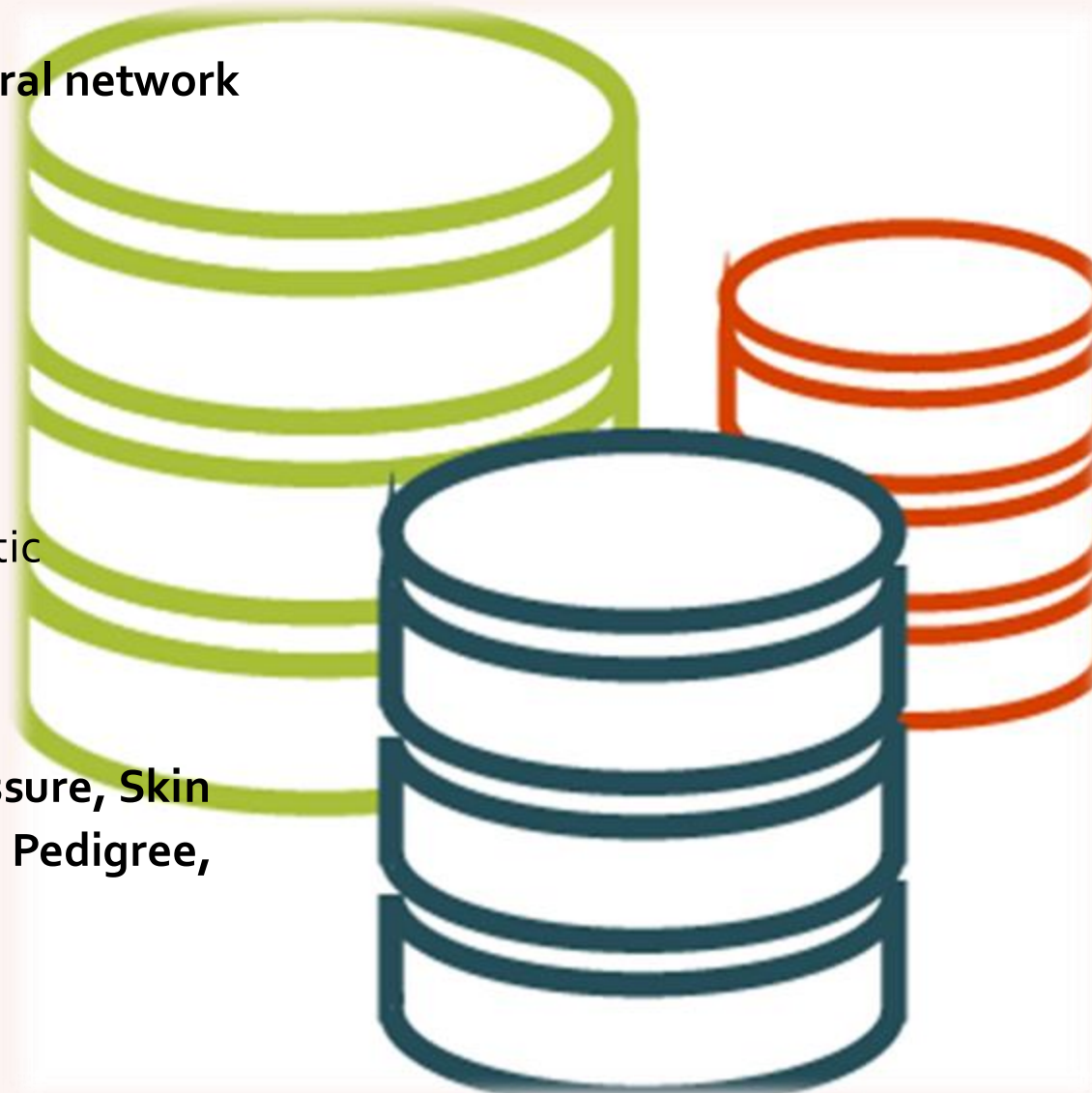
2.While maintaining:

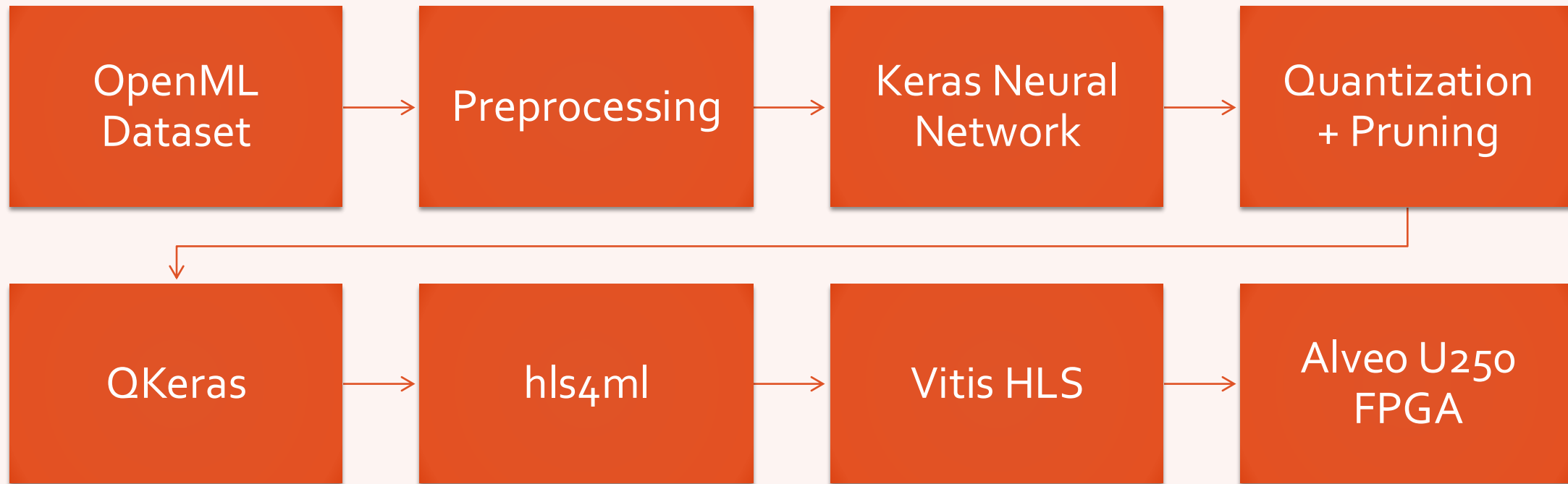
- Classification performance
- Reliable diabetes prediction

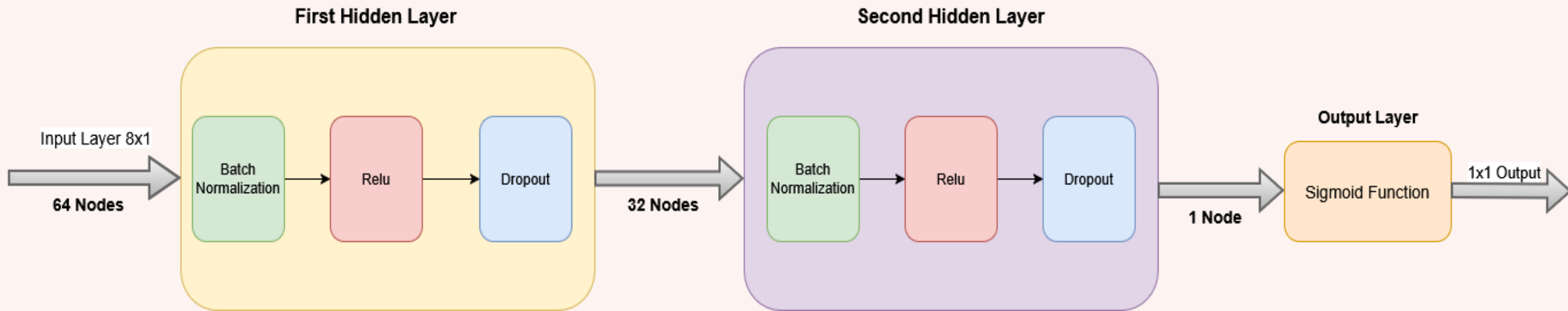
Challenges of Neural Networks on FPGAs:

- Fixed-point arithmetic
- Limited resources
- DSP/LUT consumption
- BRAM is limited

- We developed a **feed-forward neural network** for binary diabetes classification.
- **OpenML Diabetes Dataset**
 - 768 patients
 - 8 medical features
 - Binary target: diabetic / non-diabetic
- Input features
 - **Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, BMI, Diabetes Pedigree, Function, Age**







1. Quantization

FP32 → fixed point

Reduces:

- Memory
- Hardware resources

2. Pruning

Remove unimportant weights

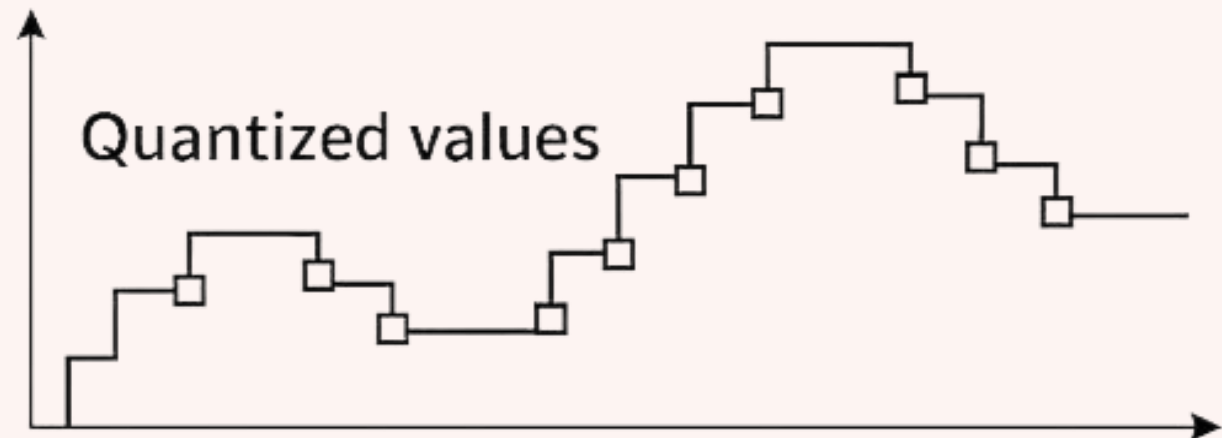
Our optimized model uses **50% sparsity**

- fewer multiplications
- fewer hardware resources

3. QAT (Quantization-Aware Training)

Instead of simply converting FP32 weights after training:

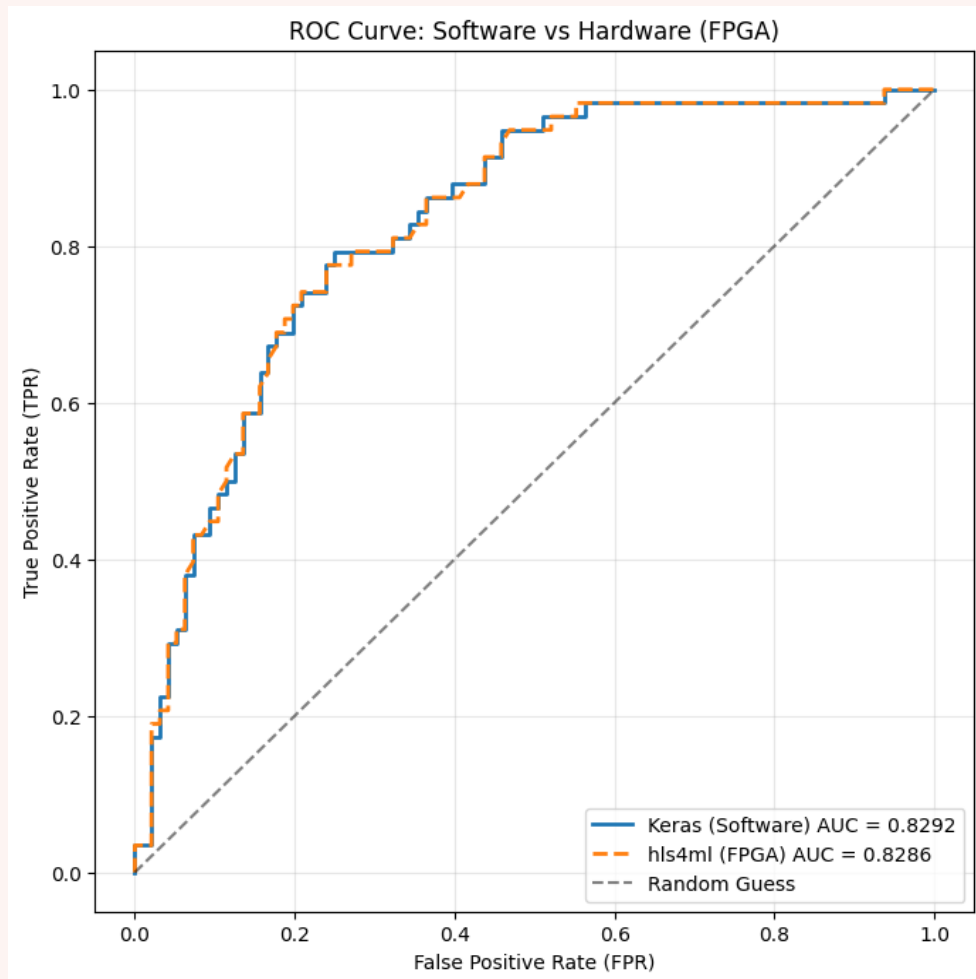
- The network learns while taking reduced precision into account.
- This helps preserve accuracy.



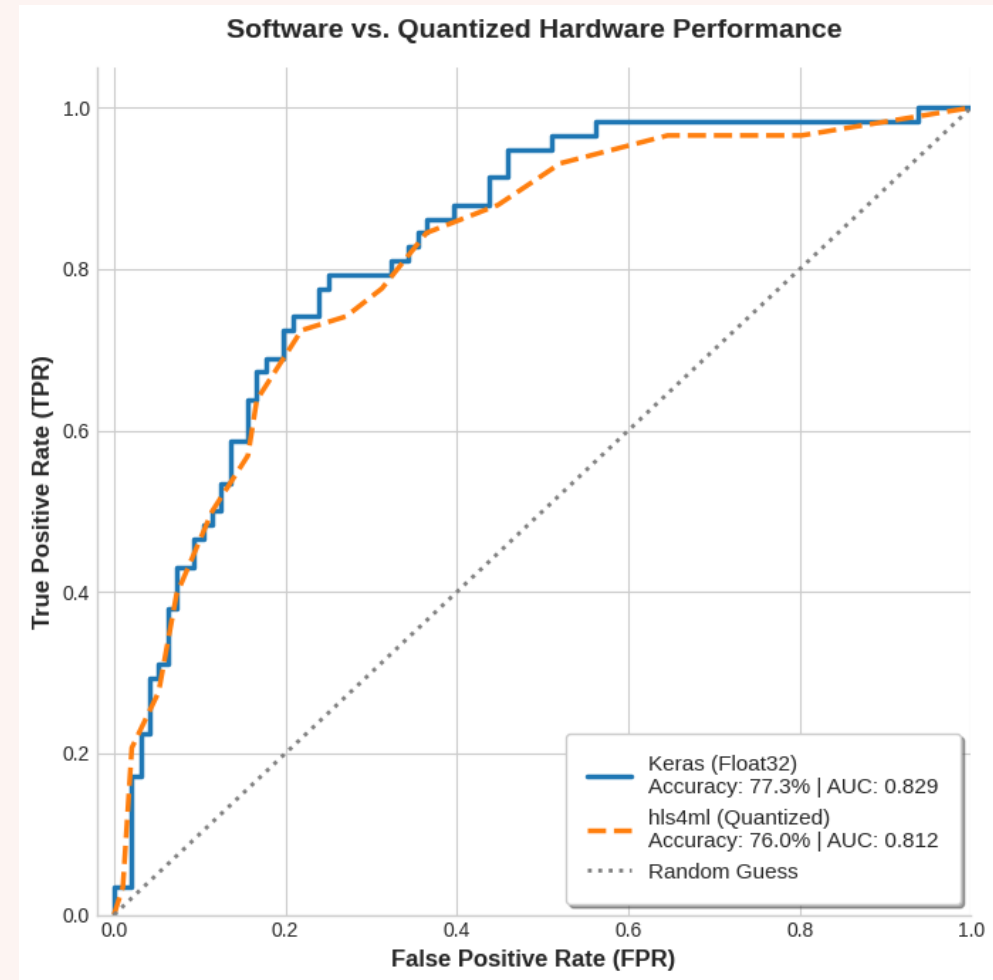
Tool	Purpose
TensorFlow / Keras	NN development & training
QKeras	Quantization-aware training
Scikit-learn	Preprocessing & evaluation
NumPy	Numerical operations
hls4ml	NN → HLS
Vitis HLS	C simulation & synthesis
Vivado	FPGA implementation & PPA

Metric	Model 1	Model 2	Model 3	Model 4
Quantized	No	Yes	No	Yes
Pruned	No	No	Yes	Yes
Clock Period	3.647 ns	3,65 ns	4,660 ns	3,108 ns
Latency	91 cycles	29 cycles	71 cycles	6 cycles
II	31	1	16	1
DSP	83	1,487	85	0
FF	24,115	110,390	35,836	1,275
LUT	73,157	48,818	39,201	12,306
BRAM	1	1	1	1
URAM	0	0	0	0

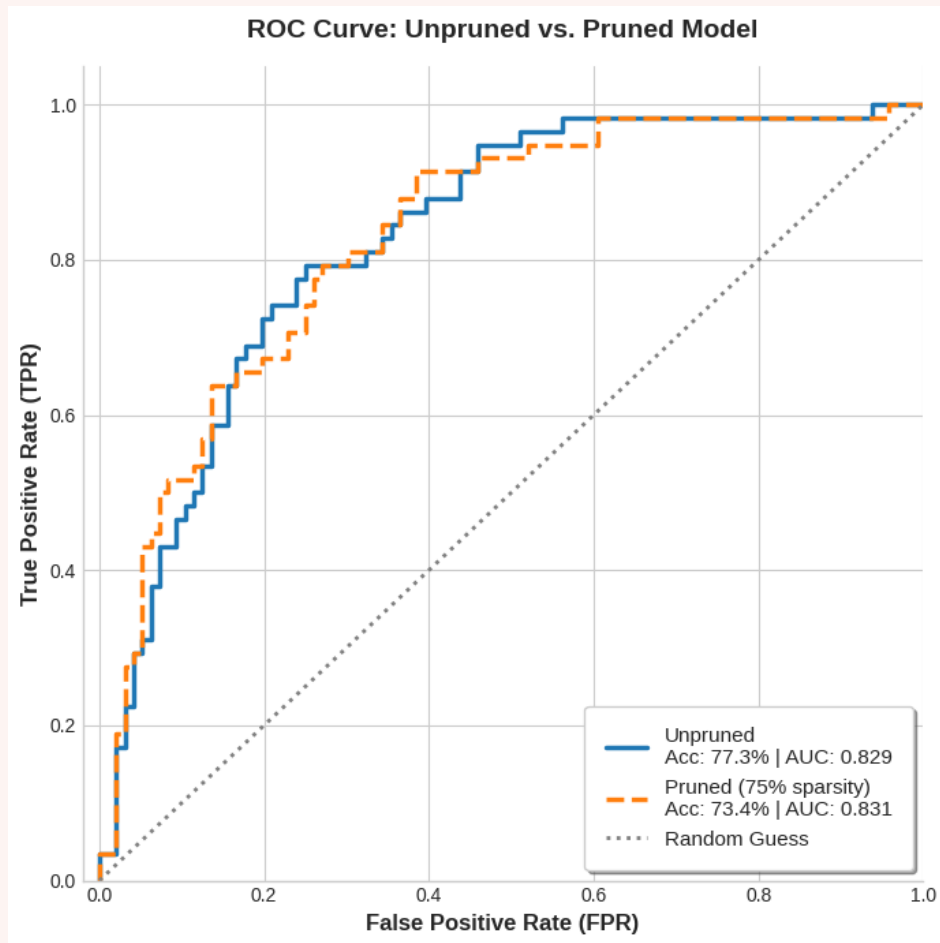
MODEL 1 (BASELINE): UNQUANTIZED AND UNPRUNED



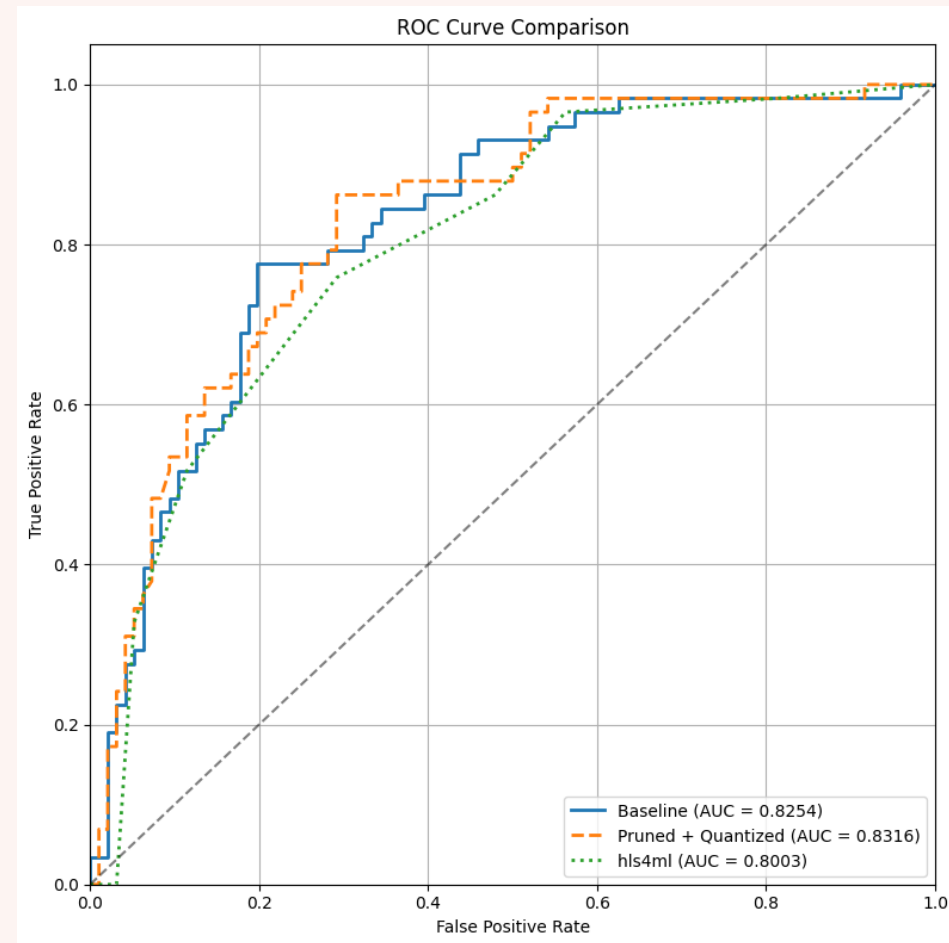
MODEL 2: QUANTIZED AND UNPRUNED.



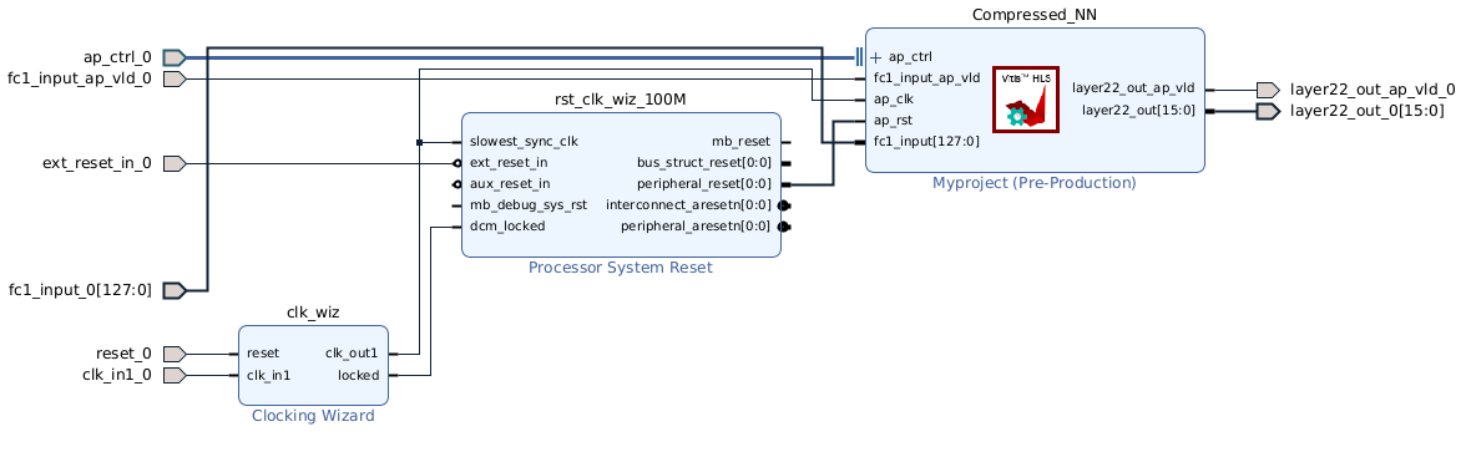
MODEL 3: UNQUANTIZED AND PRUNED.



MODEL 4: QUANTIZED AND PRUNED.



Quantization and pruning caused only a minor loss in AUC while providing major hardware benefits!!

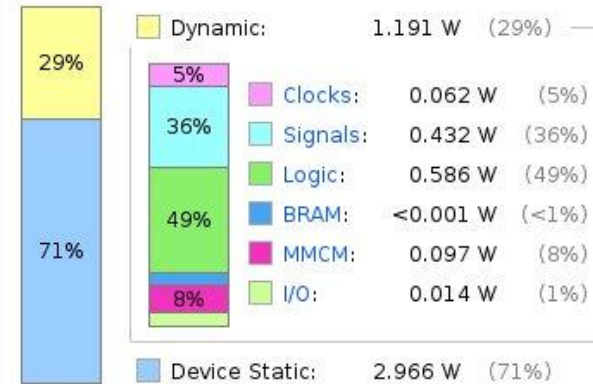


Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power: 4.157 W
Design Power Budget: Not Specified
Process: typical
Power Budget Margin: N/A
Junction Temperature: 27.2°C
 Thermal Margin: 72.8°C (126.7 W)
 Ambient Temperature: 25.0 °C
 Effective θ_{JA} : 0.5°C/W
 Power supplied to off-chip devices: 0 W
 Confidence level: Low

[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

On-Chip Power



Setup

Worst Negative Slack (WNS): 4.278 ns
 Total Negative Slack (TNS): 0.000 ns
 Number of Failing Endpoints: 0
 Total Number of Endpoints: 16549

Hold

Worst Hold Slack (WHS): 0.007 ns
 Total Hold Slack (THS): 0.000 ns
 Number of Failing Endpoints: 0
 Total Number of Endpoints: 16549

Pulse Width

Worst Pulse Width Slack (WPWS): 4.458 ns
 Total Pulse Width Negative Slack (TPWS): 0.000 ns
 Number of Failing Endpoints: 0
 Total Number of Endpoints: 10718

All user specified timing constraints are met.

	CLB LUTs (1728000)	CLB Registers (3456000)	CARRY8 (216000)	CLB (216000)	LUT as Logic (1728000)	LUT as Memory (791040)	Block RAM Tile (2688)	Bonded IOB (676)	HPIOB_M (312)	HPIOB_S (312)	HPIOB_SINGL (52)	GLOBAL CLOCK BUFFERS (1344)	MMCM (16)
1	38073	10710	1177	6398	38072	1	0.5	153	72	72	9	3	1
	38073	10710	1177	6398	38072	1	0.5	0	0	0	0	3	1

THANK YOU

tvangos@uth.gr
voikonom@uth.gr

